

$F2 \parallel WC_{\max}$ Structural Properties and Dynamic Programming

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Abstract

Minimizing $WC_{\max}$ on a two-machine flow shop, denoted $F2 \parallel WC_{\max}$, is a classical NP-hard scheduling problem. Existing exact methods rely on enumerative techniques that scale poorly with instance size, while approximation algorithms often impose restrictive assumptions on processing times. We first sequence all jobs globally by Johnson's rule, obtaining an initial schedule π . Scanning π from left to right, we partition it into at most K blocks: whenever a weight not seen before appears, it serves as the final job of the current block, and the next job starts a new block. We establish structural properties of optimal schedules: the permutation property, the positional constraint that block $\mathcal{G}_k$ jobs cannot appear beyond the first k blocks, and the block-end condition requiring the final job of each block to carry that block's weight (automatically satisfied by construction). The intra-block Johnson ordering inherited from π is further formalized within the DP framework. Building on these properties and the interval-based dynamic programming framework (Hall, 1994) developed for $F2|r_j|C_{\max}$, we design a polynomial-time dynamic programming algorithm with $O(|Y|_{\max} \cdot n)$ complexity that appends jobs block by block. We further show that geometric rounding of weights and processing times converts this algorithm into a polynomial-time approximation scheme.

Keywords: Flow shop scheduling; $WC_{\max}$; Dynamic programming; Structural properties

1 Introduction

2 Preliminaries

We consider the problem $F2 \parallel WC_{\max}$. Let $\mathcal{J} = \{J_1, J_2, \dots, J_n\}$ be the set of n jobs to be processed on two machines M_1 and M_2 in series: each job J_j must be processed first on M_1 for

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a_j time units and then on M_2 for b_j time units, with no preemption allowed. All processing times a_j and b_j are assumed to be positive integers. M_1 and M_2 can process at most one job at a time. Each job J_j has a weight $w_j > 0$. A schedule $\pi = (\pi(1), \pi(2), \dots, \pi(n))$ specifies the processing order on both machines. For a schedule π , let $A_j = \sum_{h=1}^j a_{\pi(h)}$ and $B_j = \sum_{h=1}^j b_{\pi(h)}$ be the cumulative M_1 and M_2 completion times after the first j jobs. The objective is to find a schedule π that minimizes $WC_{\max}$.

The problem $F2 \parallel WC_{\max}$ can be formulated as a mixed integer linear program (MILP). For a permutation schedule, let the positions be indexed by $k = 1, \dots, n$. We introduce the following decision variables:

- $x_{jk} = 1$ if job J_j processing at position k , and 0 otherwise;
- C_k^1 completion time of the job at position k on M_1 ;
- C_k^2 completion time of the job at position k on M_2 ;
- C_j completion time of job J_j on M_2 ;
- Z the maximum weighted completion time $WC_{\max}$.
- $M = 2 \sum_{j=1}^n (a_j + b_j)$ be a sufficiently large constant.

The MILP model is:

$$\min \quad Z \tag{1}$$

$$\text{s.t.} \quad \sum_{k=1}^n x_{jk} = 1, \quad \forall j \in \mathcal{J} \tag{2}$$

$$\sum_{j=1}^n x_{jk} = 1, \quad \forall k = 1, \dots, n \tag{3}$$

$$C_1^1 = \sum_{j=1}^n a_j x_{j1} \tag{4}$$

$$C_k^1 = C_{k-1}^1 + \sum_{j=1}^n a_j x_{jk}, \quad \forall k = 2, \dots, n \tag{5}$$

$$C_1^2 = C_1^1 + \sum_{j=1}^n b_j x_{j1} \tag{6}$$

$$C_k^2 \geq C_k^1 + \sum_{j=1}^n b_j x_{jk}, \quad \forall k = 2, \dots, n \tag{7}$$

$$C_k^2 \geq C_{k-1}^2 + \sum_{j=1}^n b_j x_{jk}, \quad \forall k = 2, \dots, n \tag{8}$$

$$C_j \geq C_k^2 - M(1 - x_{jk}), \quad \forall j \in \mathcal{J}, k = 1, \dots, n \quad (9)$$

$$Z \geq w_j C_j, \quad \forall j \in \mathcal{J} \quad (10)$$

$$x_{jk} \in \{0, 1\}, \quad C_k^1, C_k^2, C_j, Z \geq 0 \quad (11)$$

Constraint (2)–(3) ensure a one-to-one assignment between jobs and positions. Constraints (4)–(5) compute the completion time of each position on M_1 . Constraints (6)–(8) compute the completion time of each position on M_2 : for $k \geq 2$, the job at position k starts on M_2 only after it completes M_1 and the preceding job releases M_2 . Constraint (9) links the position-based completion times to the job-based completion times via a big- M formulation: when $x_{jk} = 1$, job J_j is at position k and $C_j \geq C_k^2$; when $x_{jk} = 0$, the right-hand side becomes $C_k^2 - M \leq 0$, which is non-binding. Constraint (10) enforces $Z \geq w_j C_j$ for every job, so that minimizing Z yields the optimal $WC_{\max}$.

3 Problem formulation

Theorem 3.1. $F2 \parallel WC_{\max}$ is NP-hard.

Proof. We reduce from the Partition problem: given n positive integers $x_1, x_2, \dots, x_r$ with $\sum_{i=1}^r x_i = 2T$, does there exist subset Z_1 and $Z_2 \subseteq \{1, \dots, r\}$ such that: $\sum_{i \in Z_1} x_i = \sum_{i \in Z_2} x_i = T$?

Construct an instance of $F2 \parallel WC_{\max}$. Let the number of jobs $n = r + 1$. Let $a_j = 1$, $b_j = rx_j$, $w_j = T + 1$ for jobs $j = 1, \dots, n - 1$. Put $a_n = rT$, $b_n = 1$, $w_n = 2T + 1$ for job J_n . Does there exist a threshold Y such that:

$$WC_{\max} \leq Y = r(T + 1)(2T + 1)$$

holds?

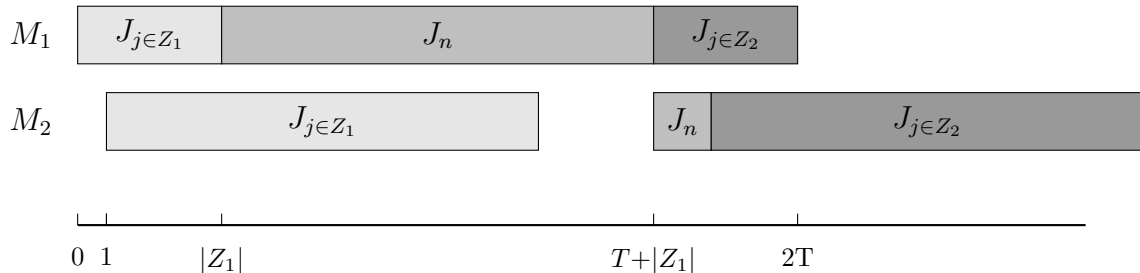


Figure 3-1

We partition jobs j_1 to j_{n-1} into two subsets via j_n , the distribution of jobs on the machine is shown in Figure 3. Since the processing times of A_1 and A_2 are unknown, while their total processing time is $2T$. We may assume that the processing time of A_1 on M_2

is $r(T + \xi)$, A_2 is $r(T - \xi)$, where $-T \leq \xi \leq T$. Therefore, the completion time of J_n is determined by the following formula:

$$C_n^1 = |A_1| + rT + 1 \quad (12)$$

$$C_n^2 = 1 + r(T + \xi) + 1 \quad (13)$$

and $C_n = \max\{C_n^1, C_n^2\}$

If $\xi = 0$, then $\sum_{j \in A_1} b_j = \sum_{j \in A_2} b_j$, which means the partition problem has a feasible solution. In this case, the completion time of job J_n is determined by (12), we have:

$$\begin{aligned} C_n &= |A_1| + rT + 1 \leq (rT + r) = r(T + 1) \\ w_n C_n &= (2T + 1)(r(T + 1)) = Y \\ WC_{\max} &= r(T + 1)(C_n + rT) = Y \end{aligned}$$

so $w_j C_j \leq WC_{\max} = Y$

If the scheduling problem exists a threshold y such that $WC_{\max} \leq y$. When $\xi > 0$, C_n is determined by (13), we have:

$$\begin{aligned} C_n &= (1 + r(T + \xi) + 1) \geq (1 + r(T + 1) + 1) \\ w_n C_n &\geq (2T + 1)(1 + r(T + 1) + 1) = Y \end{aligned}$$

When $\xi < 0$, C_n is determined by (12), we have:

$$\begin{aligned} C_{\max} &= (C_n + \sum_{j \in A_2} b_j) \\ &= (1 + rT + |A_1| + r(T - \xi)) \\ &\geq (rT + r + rT) \\ &= r(2T + 1) = Y \\ WC_{\max} &\geq r(T + 1)(r(2T + 1)) = Y \end{aligned}$$

so $WC_{\max} > Y$, there exists a contradiction.

Therefore, the Partition instance admits a solution if and only if the constructed $F2 \parallel WC_{\max}$ instance admits a schedule with $WC_{\max} \leq Y$. $\square$

Theorem 3.2. $F2 \parallel WC_{\max}$ is strongly NP-hard.

Proof. We reduce from the 3-Partition problem: given $3m$ positive integers $x_1, \dots, x_{3m}$ and a bound B with $B/4 < x_i < B/2$ and $\sum_{i=1}^{3m} x_i = mB$, can the integers be partitioned into m triples each summing to B ? Construct an instance of $F2 \parallel WC_{\max}$ with $n = 4m$ jobs as follows (see Figure 3):

- $3m$ element jobs $J_1, \dots, J_{3m}$: $a_j = 1$, $b_j = x_j + \frac{2}{3}$, $w_j = 1$;
- m separator jobs $S_1, \dots, S_m$: $a_{s_k} = B$, $b_{s_k} = 0$, $w_{s_k} = 3m/k$.

The total M_1 work is $\sum a_j = 3m + mB = m(B + 3)$, the total M_2 work is $\sum b_j = \sum_{j=1}^{3m} (x_j + \frac{2}{3}) = mB + 2m = m(B + 2)$, and the threshold is set to $Y = 3m(B + 3)$.

Suppose 3-Partition exists. There is a partition of the $3m$ elements into m triples $G_1, \dots, G_m$ with $\sum_{j \in G_k} x_j = B$ for each k . Arrange the jobs in the order $G_1, S_1, G_2, S_2, \dots, G_m, S_m$. On M_1 , the three elements of G_k each consume 1 unit (3 in total), after which S_k occupies B units; thus S_k completes M_1 at time $k(B + 3)$. On M_2 , the three elements of G_k have total processing time $\sum (x_j + \frac{2}{3}) = B + 2$. The first element of G_1 finishes M_1 at time 1 and starts M_2 immediately, finishing M_2 at $1 + (x_1 + \frac{2}{3})$. Since each $x_j > B/4$ is large relative to the unit M_1 times, the M_2 pipeline is the bottleneck, and the last element of G_1 completes M_2 at $B + 3$. S_1 has $b = 0$ and passes M_2 instantly at $B + 3$. Inductively, S_k completes M_2 at $C_{s_k} = k(B + 3)$. Consequently

$$w_{s_k} C_{s_k} = \frac{3m}{k} \cdot k(B + 3) = 3m(B + 3) = Y,$$

while each element job has $w_j = 1$ and completion time at most $m(B + 3)$, giving $w_j C_j \leq m(B + 3) \leq Y$ for all $m \geq 3$. Hence $WC_{\max} = Y$.

Conversely, suppose there exists a schedule π with $WC_{\max} \leq Y = 3m(B + 3)$. For separator S_k we have $w_{s_k} C_{s_k} = \frac{3m}{k} C_{s_k} \leq 3m(B + 3)$, i.e., $C_{s_k} \leq k(B + 3)$. Because each separator requires B time on M_1 , the k -th separator in schedule order cannot finish M_2 before M_1 has processed at least k separators plus all element jobs preceding them. Let E_k be the number of element jobs preceding S_k , and let $e_k = E_k - E_{k-1}$ (with $E_0 = 0$) be the number of element jobs between S_{k-1} and S_k . From $C_{s_k} \geq kB + E_k$ and $C_{s_k} \leq k(B + 3)$, we obtain $E_k \leq 3k$. Since $E_m = 3m$, each inequality must hold with equality: $E_k = 3k$ for all k , hence $e_k = 3$ for every block. Denote the three elements between S_{k-1} and S_k by G_k , with $\sum_{j \in G_k} x_j = B_k$.

Now analyze the M_2 pipeline of block k . The three elements of G_k have total M_2 work $B_k + 2$ and unit M_1 time each. From the flow-shop critical-path formula applied to G_k followed by S_k , the span satisfies $C_{s_k} \geq C_{s_{k-1}} + \max\{B_k + 3, B + 3\}$ (the $B_k + 3$ term arises from the first element starting M_2 one unit after the block begins, followed by $B_k + 2$ total M_2 work on the three elements of G_k , while $B + 3$ is the pure M_1 path through S_k). Since $C_{s_k} \leq C_{s_{k-1}} + B + 3$ (otherwise C_{s_k} exceeds $k(B + 3)$ given $C_{s_{k-1}} \leq (k - 1)(B + 3)$), we obtain

$$B_k + 3 \leq C_{s_k} - C_{s_{k-1}} \leq B + 3 \implies B_k \leq B.$$

Summing over all m groups yields $\sum_{k=1}^m B_k \leq mB$, but the total of all x_j equals mB by construction; therefore each inequality holds with equality: $\sum_{j \in G_k} x_j = B$ for every k . With $B/4 < x_j < B/2$, three elements can sum to exactly B only if each group is a valid

triple; fewer than three would sum to $< B$, more than three to $> B$. Hence $\{G_1, \dots, G_m\}$ is a partition of the $3m$ elements into m triples each summing to B .

Thus $F2 \parallel WC_{\max}$ is strongly NP-hard. $\square$

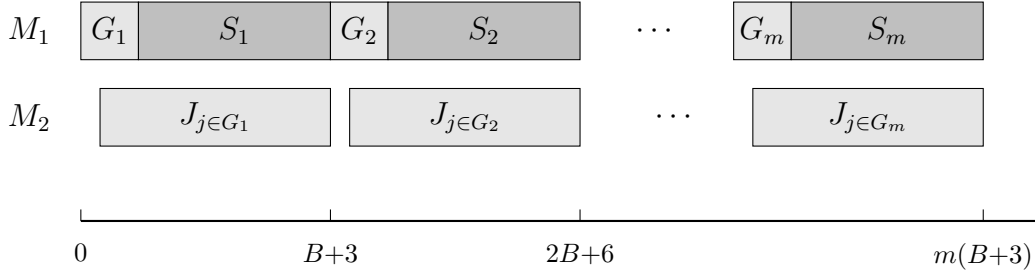


Figure 3-2: 3-Partition reduction;

4 Algorithm

In this chapter, we will propose an idea to guide the dynamic programming algorithm for solving this problem, and present the time complexity analysis of the algorithm.

4.1 The number of weights is constant

For $F2 \parallel WC_{\max}$, a permutation schedule in which the job order on M_1 and M_2 , Possess an identical processing order for jobs (Baker, 1974). Johnson's rule (Johnson, 1954) sequences all n jobs to minimize the makespan: jobs with $a_j \leq b_j$ go first in non-decreasing order of a_j , and the remaining jobs follow in non-increasing order of b_j .

For any scheduling π . Let $W_1, W_2, \dots, W_K$ be the K distinct weights among all jobs, and $W_1 > W_2 > \dots > W_K$ and define $\mathcal{G}_k = \{J_j \in \mathcal{J} \mid w_j = W_k\}$, $k = 1, \dots, K$. For each group $\mathcal{G}_k$, let $\ell_k = \arg \max\{p : \pi(p) \in \mathcal{G}_k\}$, $p \in \{1, \dots, n\}$, and set $\ell_0 = 0$; The blocks are defined as: For $J_j \in \mathcal{J}$, $\mathcal{B}_1 = \{J_1, \dots, J_{\ell_1}\}$, $\mathcal{J} = \{\mathcal{J} \setminus \mathcal{B}_1\}$; $\mathcal{B}_2 = \{J_{\ell_1}, \dots, J_{\ell_2}\}$, $\mathcal{J} = \{\mathcal{J} \setminus \mathcal{B}_2\}$; $\mathcal{B}_k = \{J_{\ell_{k-1}}, \dots, J_{\ell_k}\}$.

Note a block is allowed empty and accepts any job. In particular, if $\ell_k = \ell_{k-1}$, then $\mathcal{B}_k = \{J_{\ell_{k-1}}, \dots, J_{\ell_k}\}$ contains no job and is called an empty block; an empty block $\mathcal{B}_i$ has $A_i = B_i = C_i = 0$.

A_i total processing time on machine M_1 of jobs in block $\mathcal{B}_i$; B_i total processing time on machine M_2 of jobs in block $\mathcal{B}_i$; C_i completion time of block $\mathcal{B}_i$ on machine M_2 ; L_i weight of the last job in block $\mathcal{B}_i$. Thus each block weights are non-increasing: $W_{\mathcal{B}_1} \geq W_{\mathcal{B}_2} \geq \dots \geq W_{\mathcal{B}_K}$.

For ease of explanation, we construct an illustrative schedule.

Example 4.1. Consider a 6-job instance with sequence $\pi = (J_1, \dots, J_6)$ and the following (a_j, b_j, w_j) values:

	J_1	J_2	J_3	J_4	J_5	J_6
a_j	2	5	6	8	4	3
b_j	5	7	9	4	3	2
w_j	2	2	5	2	4	2

The resulting are $\mathcal{G}_1 = \{J_3\}$, $\mathcal{G}_2 = \{J_5\}$, $\mathcal{G}_3 = \{J_1, J_2, J_4, J_6\}$. And $\mathcal{B}_1 = \{J_1, J_2, J_3\}$, $\mathcal{B}_2 = \{J_4, J_5\}$, $\mathcal{B}_3 = \{J_6\}$.

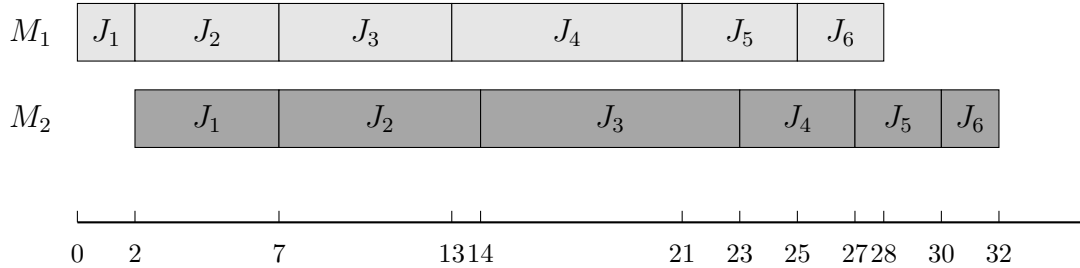


Figure 4.1

Lemma 4.1. For any schedule π , reordering the jobs within each block according to Johnson's rule Johnson, 1954 and concatenating the resulting block schedules yields a schedule π' with $WC_{\max}(\pi') \leq WC_{\max}(\pi)$.

Proof. By construction, the final job of block $\mathcal{B}_k$ carries weight $W(\mathcal{B}_k)$, and every job in $\mathcal{B}_k$ has weight at most $W(\mathcal{B}_k)$: a weight occurring in $\mathcal{B}_k$ either equals $W(\mathcal{B}_k)$ or has already appeared in an earlier block with larger weight. Let $C_k(\sigma)$ denote the completion time of the last job of $\mathcal{B}_k$ under schedule σ .

We first argue that within each block, the maximum weighted completion time is attained at its last job. For any $J_j \in \mathcal{B}_k$, we have $C_j(\sigma) \leq C_k(\sigma)$ because the last job finishes no earlier than any other job in the block, and $w_j \leq W(\mathcal{B}_k)$ by the block construction. Hence $w_j C_j(\sigma) \leq W(\mathcal{B}_k) \cdot C_k(\sigma)$. Taking the maximum over $j \in \mathcal{B}_k$ yields $\max_{j \in \mathcal{B}_k} w_j C_j(\sigma) = W(\mathcal{B}_k) \cdot C_k(\sigma)$, and therefore

$$WC_{\max}(\sigma) = \max_{k=1, \dots, K} \{W(\mathcal{B}_k) \cdot C_k(\sigma)\}.$$

Thus the overall objective depends only on the last job of each block.

Now let π' be the schedule obtained by reordering each $\mathcal{B}_k$ internally via Johnson's rule, preserving the block order. We prove $C_k(\pi') \leq C_k(\pi)$ for all k by induction.

Base case $k = 1$. Block $\mathcal{B}_1$ starts at time zero on both machines regardless of internal ordering, so Johnson's rule directly gives $C_1(\pi') \leq C_1(\pi)$.

Inductive step. Assume $C_{k-1}(\pi') \leq C_{k-1}(\pi)$. On M_1 , the start time of $\mathcal{B}_k$ equals the total M_1 work of $\mathcal{B}_1, \dots, \mathcal{B}_{k-1}$, which is invariant under internal reordering; thus $\mathcal{B}_k$ begins at the same instant on M_1 under both schedules. On M_2 , $\mathcal{B}_k$ starts at $\max\{\sum_{i=1}^k A_i, C_{k-1}\}$. The first term is identical under π and π' , and the second is no larger under π' by the induction hypothesis. Hence $\mathcal{B}_k$ starts no later on M_2 under π' than under π . With the same M_1 start and an internally optimal Johnson ordering, we obtain $C_k(\pi') \leq C_k(\pi)$, completing the induction.

Finally,

$$WC_{\max}(\pi') = \max_k W(\mathcal{B}_k) \cdot C_k(\pi') \leq \max_k W(\mathcal{B}_k) \cdot C_k(\pi) = WC_{\max}(\pi),$$

which establishes the lemma. $\square$

4.2 Algorithm Properties and Dynamic Programming

We now develop a dynamic programming algorithm that processes jobs sequentially in Johnson order and assigns each job to one of the K blocks $\mathcal{B}_1, \dots, \mathcal{B}_K$.

Lemma 4.1 provides the structural basis: it guarantees that, given any assignment of jobs to the K blocks, sequencing the jobs within each block $\mathcal{B}_i$ according to Johnson's rule and concatenating the blocks in non-increasing weight order yields a schedule whose makespan is no larger than that of any other schedule respecting the same assignment. For each job $J_j \in \mathcal{J}$ ($j = 1, \dots, n$), let

$$\mathcal{H}_j = \{A_i; B_i, C_i; L_i\}$$

denote the state associated with J_j , where the index i ranges over the blocks eligible for J_j , and in every block preceding it:

- A_i — total processing time on machine M_1 of jobs in block $\mathcal{B}_i$;
- B_i — total processing time on machine M_2 of jobs in block $\mathcal{B}_i$;
- C_i — completion time of the last job of block $\mathcal{B}_i$ on machine M_2 ;
- L_i — weight of block $\mathcal{B}_i$.

Since the block weights are non-increasing, $W_1 > \dots > W_K$, the blocks eligible for J_j form a prefix:

$$\mathcal{E}(J_j) = \{i : w_j \leq W_i\}, \quad i^* = \arg \max_{i \in \{1, \dots, K\}} \{i : W_i \geq w_j\},$$

so that $\mathcal{E}(J_j) = \{1, \dots, i^*\}$. A state is in fact the aggregate K -block tuple, described by $4K - 2$ parameters: A_K and B_K need not be stored, since after the first j jobs have been

processed, every job belongs to exactly one block and the total processing times $\sum_{i=1}^K A_i = \sum_{j=1}^n a_{\pi(j)}$ and $\sum_{i=1}^K B_i = \sum_{j=1}^n b_{\pi(j)}$ are known, so A_K and B_K are recovered by

$$A_K = \sum_{j=1}^n a_{\pi(j)} - \sum_{i=1}^{K-1} A_i, \quad B_K = \sum_{j=1}^n b_{\pi(j)} - \sum_{i=1}^{K-1} B_i.$$

Hence a state is represented as

$$(C_1, \dots, C_K; A_1, \dots, A_{K-1}; B_1, \dots, B_{K-1}; L_1, \dots, L_K) \in \mathcal{H}_j.$$

The initial state set contains a single state with all components set to zero: $\mathcal{H}_0 = \{(0, \dots, 0; 0, \dots, 0; 0, \dots, 0; 0, \dots, 0)\}$.

Lemma 4.2. *Let $H = (C_1, \dots, C_K; A_1, \dots, A_{K-1}; B_1, \dots, B_{K-1}; L_1, \dots, L_K)$ and $H' = (C'_1, \dots, C'_K; A'_1, \dots, A'_K; B'_1, \dots, B'_K; L'_1, \dots, L'_K)$ be two states in $\mathcal{H}_j$ for some j . If $A_i = A'_i$, $B_i = B'_i$, $L_i = L'_i$ for all i , and $C_i \leq C'_i$ for all i , then H' can be discarded without affecting the optimal solution.*

Proof. Consider any continuation of a schedule extending state H' to a complete schedule Π' . Replacing the prefix represented by H' with the prefix represented by H yields a schedule Π that uses the same assignment of the remaining jobs. For every subsequent block $\mathcal{B}_k$ ($k \geq 2$), the M_2 start time is $\max\{\sum_{i=1}^k A_i, C_{k-1}\}$, while C_{k-1} under H is no larger than under H' . By induction, the M_2 completion time of every subsequent block is no larger under H , and since A_i, B_i, L_i are identical, the objective value $WC_{\max}$ of Π does not exceed that of Π' . $\square$

During the merge step, the algorithm applies this dominance rule: among states with identical $(A_i, B_i, L_i)_{i=1}^K$, only those with component-wise minimal $(C_i)_{i=1}^K$ are retained.

For $j = 1, \dots, n$, the algorithm extends $\mathcal{H}_{j-1}$ to $\mathcal{H}_j$ as follows. For each state in $\mathcal{H}_{j-1}$ and each block $i \in \mathcal{E}(J_j)$ (every block that precedes the block of J_j , i.e., every i with $w_j \leq L_i$), the algorithm places J_j at the end of $\mathcal{B}_i$. Only the i -th slice of the state is updated, while all other blocks remain unchanged:

$$C'_i = \max\{C_i + b_j, A_i + a_j + b_j\}, \quad A'_i = A_i + a_j, \quad B'_i = B_i + b_j.$$

The resulting candidate states are collected in a temporary set $\mathcal{Y}$ (the sub-states). Via Lemma 4.2, merging the candidates removes dominated and duplicate states and yields $\mathcal{H}_j$.

Once all jobs are assigned and each block $\mathcal{B}_i$ is summarized by its parameters (A_i, B_i, C_i) , the overall makespan $C_{\max}$ is obtained by concatenating the blocks in order of increasing index. The procedure is given below:

In the update of T_B , the term $T_A - A_i$ is the start time of $\mathcal{B}_i$ on M_1 , so $T_A - A_i + C_i$ is the completion time of $\mathcal{B}_i$ on M_2 when M_2 is free, while $T_B + B_i$ accounts for the case where M_2 is the bottleneck.

Algorithm 1 Block concatenation and evaluation

1: **Input:** States $\mathcal{H}$.
2: **Output:** $WC_{\max} = \max_i L_i \cdot T_B^{(i)}$.
3: $T_{A_0} \leftarrow 0, T_{B_0} \leftarrow 0, WC_{\max} \leftarrow 0$
4: **for** $i = 1$ to K **do**
5: $T_{A_i} \leftarrow T_{A_{i-1}} + A_i$
6: $T_{B_i} \leftarrow \max\{T_{A_{i-1}} + C_i, T_{B_{i-1}} + B_i\}$
7: $WC_{\max} \leftarrow \max\{WC_{\max}, L_i \cdot T_{B_i}\}$
8: **end for**
9: **return** $WC_{\max}$

Algorithm 2 DP for $F2 \parallel WC_{\max}$

1: **Input:** Jobs $\mathcal{J}$ with (a_j, b_j, w_j) in Johnson order; block count K ; distinct weights $W_1 > \dots > W_K$.
2: **Output:** The optimal value $Z^* = \min_S \max_i W_i \cdot T_B^{(i)}$.
3: Set $L_i \leftarrow W_i$ for $i = 1, \dots, K$
4: $\mathcal{H}_0 \leftarrow \{(0, \dots, 0; 0, \dots, 0; 0, \dots, 0; 0, \dots, 0)\}$
5: **for** $j = 1$ to n **do**
6: $\mathcal{Y}_i \leftarrow \emptyset$ for $i = 1, \dots, K$
7: **for each** $(\{C_i, A_i, B_i, L_i\}_{i=1}^K) \in \mathcal{H}_{j-1}$ **do**
8: **for** $i = 1$ to K **do**
9: **if** $w_j \leq L_i$ **then**
10: $C'_i \leftarrow \max\{C_i + b_j, A_i + a_j + b_j\}$
11: $A'_i \leftarrow A_i + a_j, B'_i \leftarrow B_i + b_j$
12: Add the updated state to $\mathcal{Y}_i$
13: **end if**
14: **end for**
15: **end for**
16: $\mathcal{H}_j \leftarrow \text{MERGE}(\mathcal{Y}_1, \dots, \mathcal{Y}_K)$
17: **end for**
18: $Z^* \leftarrow +\infty$
19: **for each state** $S = \{C_i, A_i, B_i, L_i\}_{i=1}^K \in \mathcal{H}_n$ **do**
20: $WC_{\max} \leftarrow \text{ALGORITHM 1}((A_i, B_i, C_i, L_i)_{i=1}^K)$
21: $Z^* \leftarrow \min\{Z^*, WC_{\max}\}$
22: **end for**
23: **return** Z^*

Theorem 4.1. *Algorithm 2 solves $F2 \parallel WC_{\max}$ optimally in $O(n \cdot K^2 \cdot |\mathcal{H}|_{\max})$ time, where $|\mathcal{H}|_{\max} \leq K^K \cdot V^{3K-3}$, $V = \max\{\sum_{j=1}^n a_j, \sum_{j=1}^n b_j\}$. For constant K , this is $O(n \cdot V^{3K-3})$.*

Proof. Each state in $\mathcal{H}_j$ is a $(4K-2)$ -tuple $(C_1, \dots, C_K; A_1, \dots, A_{K-1}; B_1, \dots, B_{K-1}; L_1, \dots, L_K)$.

We bound the number of distinct values each component can attain.

The quantity A_i is the total M_1 processing time of jobs assigned to block $\mathcal{B}_i$, hence $A_i \in [0, V]$. Since the job sequence is fixed and the j jobs under consideration form a prefix of the Johnson order, $\sum_{i=1}^K A_i$ equals the sum of a_j over that prefix. Consequently at most $K - 1$ of the A_i are independent, contributing at most V^{K-1} distinct A -vectors. Symmetrically, $\sum_{i=1}^K B_i$ is the prefix sum of the b_j 's, so the B_i components likewise contribute at most V^{K-1} distinct combinations.

The quantity C_i is the makespan of $\mathcal{B}_i$ when scheduled in isolation by Johnson's rule. Since $C_i \leq A_i + B_i \leq 2V$, the C -vector contributes at most $(2V)^K$ distinct values. The weight L_i is the weight of the last job in $\mathcal{B}_i$ and takes values from $\{w^{(1)}, \dots, w^{(K)}\}$, yielding at most K^K distinct L -vectors.

Multiplying these bounds yields

$$|\mathcal{H}_j| \leq K^K \cdot (2V)^{K-1} \cdot V^{2K-2} = O(K^K \cdot 2^K \cdot V^{3K-3}).$$

In iteration j , the algorithm takes each state in $\mathcal{H}_{j-1}$, tries every block $i \in \{1, \dots, K\}$, computes the updated tuple in $O(1)$ time, and inserts it into Y_i . The merge step collects the K sets $Y_1, \dots, Y_K$ and removes duplicates in time proportional to the total number of generated tuples. Hence iteration j costs $O(K \cdot |\mathcal{H}_{j-1}|)$, and summing over all n iterations gives $O(n \cdot K \cdot |\mathcal{H}|_{\max})$.

The final concatenation via Algorithm 1 evaluates each state in $\mathcal{H}_n$ in $O(K)$ time, which is dominated by the DP phase. When K is fixed, $K^K \cdot 2^K$ is constant, giving $O(n \cdot V^{3K-3})$. $\square$

4.3 FPTAS

We turn the exact dynamic program of Algorithm 2 into a fully polynomial-time approximation scheme by geometric rounding of its state space. Only the completion-time coordinates A_i, B_i, C_i are rounded.

Fix $0 < \varepsilon < 1$ and let

$$\delta = 1 + \frac{\varepsilon}{2n}, \quad U = 2V, \quad v = \left\lceil \log_\delta U \right\rceil,$$

where $V = \max\{\sum_{j=1}^n a_j, \sum_{j=1}^n b_j\}$. Partition $[0, U]$ into the $v + 1$ intervals

$$I_0 = [0, 1), \quad I_r = [\delta^{r-1}, \delta^r) \quad (1 \leq r \leq v-1), \quad I_v = [\delta^{v-1}, U]. \quad (14)$$

All processing times are assumed integral, so every value taken by a coordinate A_i, B_i, C_i is a nonnegative integer; hence any two values lying in a common interval satisfy $x \leq \delta y$. We call this the *interval-dominance* property; it is the only fact about the rounding used below.

A *box* is the set of states whose time coordinates fall in a fixed tuple of intervals and whose weight labels are fixed. For $\mathbf{r}^A, \mathbf{r}^B, \mathbf{r}^C \in \{0, \dots, v\}^K$ and $\mathbf{L} = (L_1, \dots, L_K) \in (\{0\} \cup$

$\{W_1, \dots, W_K\}\}^K$, where $L_i = 0$ marks an empty block $\mathcal{B}_i$, the box is

$$\mathcal{B}(\mathbf{r}^A; \mathbf{r}^B; \mathbf{r}^C; \mathbf{L}) = \left\{ (C_i, A_i, B_i, L_i)_{i=1}^K : A_i \in I_{r_i^A}, B_i \in I_{r_i^B}, C_i \in I_{r_i^C} \right\}.$$

There are $(v+1)^{3K}(K+1)^K$ boxes, a number independent of the job sizes.

Algorithm 3 Boxed FPTAS for $F2 \parallel WC_{\max}$

- 1: **Input:** Jobs $\mathcal{J}$ with (a_j, b_j, w_j) in Johnson order; block count K ; accuracy ε .
 - 2: **Output:** A schedule with $WC_{\max} \leq (1 + \varepsilon) WC_{\max}^*$.
 - 3: Set $\delta = 1 + \varepsilon/(2n)$, $U = 2V$, $v = \lceil \log_\delta U \rceil$, and the intervals of (14)
 - 4: Store the zero state as the representative of $\mathcal{B}(\mathbf{0}; \mathbf{0}; \mathbf{0}; \mathbf{0})$
 - 5: **for** $j = 0$ to $n - 1$ **do**
 - 6: **for** each box $\mathcal{B}$ with representative $s = (C_i, A_i, B_i, L_i)_{i=1}^K$ **do**
 - 7: **for** $i = 1$ to K **do**
 - 8: **if** block i is empty **or** $w_{j+1} \leq L_i$ **then**
 - 9: Compute the candidate $\tilde{s}$ by the recurrences of Algorithm 2 (append J_{j+1} to block i ; if block i was empty, set $L_i \leftarrow w_{j+1}$)
 - 10: Map $\tilde{s}$ to its box $\mathcal{B}'$
 - 11: **if** $\mathcal{B}'$ has no representative **then** Store $\tilde{s}$ and record s as its predecessor
 - 12: **else** Keep either one of the two representatives of $\mathcal{B}'$
 - 13: **end if**
 - 14: **end if**
 - 15: **end for**
 - 16: **end for**
 - 17: **end for**
 - 18: Evaluate every representative at level n with Algorithm 1
 - 19: **return** the schedule attaining the minimum $WC_{\max}$
-

Theorem 4.2. *For every $0 < \varepsilon < 1$, Algorithm 3 returns a schedule with*

$$WC_{\max} \leq (1 + \varepsilon) WC_{\max}^*,$$

where $WC_{\max}^*$ is the optimal value of $F2 \parallel WC_{\max}$.

Proof. Let $s_0^* \rightarrow s_1^* \rightarrow \dots \rightarrow s_n^*$ be a trajectory of Algorithm 2 that ends in an optimal schedule, and write $s_j^* = (C_i^{j*}, A_i^{j*}, B_i^{j*}, L_i^{j*})_{i=1}^K$. We show by induction on j that, after processing j jobs, Algorithm 3 stores a representative $\hat{s}_j = (\hat{C}_i, \hat{A}_i, \hat{B}_i, \hat{L}_i)_{i=1}^K$ with $\hat{L}_i = L_i^{j*}$ for all i and

$$\hat{A}_i \leq A_i^{j*} \delta^j, \quad \hat{B}_i \leq B_i^{j*} \delta^j, \quad \hat{C}_i \leq C_i^{j*} \delta^j \quad (1 \leq i \leq K). \quad (\star)$$

The base case $j = 0$ is the stored zero state. Assume $(\star)$ holds for j , and let i^* be the block to which s_j^* appends J_{j+1} . Because the weight labels are stored exactly, block i^* is empty in

$\hat{s}_j$ whenever it is empty in s_j^* , and otherwise $w_{j+1} \leq L_{i^*}^{j*} = \hat{L}_{i^*}$; hence the same transition is applicable from $\hat{s}_j$. Applying it yields a candidate $\tilde{s}_{j+1}$ with $\tilde{L}_i = L_i^{j+1,*}$ for all i and, by the recurrences of Algorithm 2 and $\delta^j \geq 1$,

$$\begin{aligned}\tilde{A}_{i^*} &= \hat{A}_{i^*} + a_{j+1} \leq (A_{i^*}^{j*} + a_{j+1})\delta^j = A_{i^*}^{j+1,*}\delta^j, \\ \tilde{B}_{i^*} &= \hat{B}_{i^*} + b_{j+1} \leq (B_{i^*}^{j*} + b_{j+1})\delta^j = B_{i^*}^{j+1,*}\delta^j, \\ \tilde{C}_{i^*} &= \max\{\hat{C}_{i^*} + b_{j+1}, \hat{A}_{i^*} + a_{j+1} + b_{j+1}\} \\ &\leq \max\{C_{i^*}^{j*} + b_{j+1}, A_{i^*}^{j*} + a_{j+1} + b_{j+1}\}\delta^j = C_{i^*}^{j+1,*}\delta^j,\end{aligned}$$

while all other coordinates are unchanged and satisfy the bound at $j+1$. The candidate $\tilde{s}_{j+1}$ is mapped to some box $\mathcal{B}'$. By interval dominance, the representative $\hat{s}_{j+1}$ ultimately stored in $\mathcal{B}'$ satisfies $\hat{x}_i \leq \delta \tilde{x}_i$ for every $x \in \{A, B, C\}$ and every i , which yields $(\star)$ at $j+1$.

It remains to compare the objective values. In the block-concatenation evaluation of Algorithm 1, the quantity $T_{B,i}$ is nondecreasing in every coordinate and positively homogeneous, so $T_{B,i}(\hat{s}_n) \leq \delta^n T_{B,i}(s_n^*)$. With $\hat{L}_i = L_i^{n,*}$,

$$WC_{\max}(\hat{s}_n) = \max_{1 \leq i \leq K} \hat{L}_i T_{B,i}(\hat{s}_n) \leq \delta^n \max_{1 \leq i \leq K} L_i^{n,*} T_{B,i}(s_n^*) = \delta^n WC_{\max}^*.$$

Finally, $\delta^n = (1 + \varepsilon/(2n))^n \leq 1 + \varepsilon$ for $0 < \varepsilon < 1$. The schedule returned by Algorithm 3 is no worse than the schedule realized by $\hat{s}_n$, so its value is at most $(1 + \varepsilon) WC_{\max}^*$. $\square$

Theorem 4.3. *For fixed K , Algorithm 3 runs in*

$$O\left(n^{3K+1} \frac{(\log V)^{3K}}{\varepsilon^{3K}}\right)$$

time and is therefore a fully polynomial-time approximation scheme for $F2 \parallel WC_{\max}$.

Proof. At each level j , Algorithm 3 stores at most one representative per box, and there are $(v+1)^{3K}(K+1)^K$ boxes. Each representative generates K candidates in $O(1)$ time, and the final evaluation of the representatives at level n costs $O(K)$ per box. Hence the running time is

$$O(nK(v+1)^{3K}(K+1)^K).$$

Since $\ln(1 + \varepsilon/(2n)) \geq \varepsilon/(4n)$ for $0 < \varepsilon < 1$,

$$v = \left\lceil \frac{\ln U}{\ln \delta} \right\rceil = O\left(\frac{n \log V}{\varepsilon}\right),$$

and for constant K the bound reduces to $O(n^{3K+1}(\log V)^{3K}/\varepsilon^{3K})$, polynomial in n , $\log V$, and $1/\varepsilon$. $\square$

5 Conclusion

References

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